

CQAs Challenges and Impact on End-To-End Integrated Continuous Biomanufacturing

Robert Dream

*Managing Director
HDR Company LLC*

Introduction

Whether you have implemented the full end-to-end Integrated Continuous Biomanufacturing (ICB), or a hybrid version of the technology, and/or are considering it in one way or another, continuous bioprocessing has become a household word among biopharma companies. Continuous manufacturing has the potential to increase the efficiency, flexibility, agility and robustness of manufacturing by reducing the number of steps and holds, utilizing smaller equipment and facilities, improving product quality and enabling real-time release. To realize the real time release of products; we have to have all the requirements in place that is needed for real time release, integrated into the manufacturing operation to assure that the product quality, safety, and efficacy is within the licensed BLA limits of the product CQAs and deliver a product that is in compliance with all the requirement hence set forth.

Controlling Continuous Manufacturing via CQAs and QbD

To produce viable drug products on an end-to-end continuous manufacturing setting, an adherence to the regulatory compliance requirement is needed. Biological medicinal therapeutic products such as monoclonal antibodies (mAbs) are particularly complex and can have numerous quality attributes that can potentially impact safety and/or efficacy of the drug product. Identifying Critical Quality Attributes (CQAs) for a biological therapeutic drug substances

and drug products are arguably the most difficult step in the implementation of Quality and Design (QbD) for development and production of biopharmaceuticals. CQAs are defined as “a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality”.¹

Structural characterization is used to assess CQAs of biopharmaceutical substance and product. The structural data must be supported by functional data to establish a structure-function relationship. These data could then be used to define the structural components' impact on the activity of the product. The characterization data obtained are essential for product development and regulatory acceptance. Characterization of multiple product batches in a continuous setting operation is essential to demonstrate that the manufacturer has control of the manufacturing process by analyzing a number of batches of the product and comparing the data. Significant differences between batches in a continuous operation setting need to be investigated and their impact on the function of the product assessed. This comparison in the QbD paradigm also centers on the CQAs (Figure 1).

The key differentiators of QbD is an enhanced, Quality by Design approach to product development in a continuous manufacturing would additionally include the following: Systematic evaluation, understanding and refining of the formulation and manufacturing process, including, identifying prior knowledge, experimentation, risk assessment, the material attributes, and process parameters that can have an effect on product CQAs. Determining the functional relationships that link critical material attributes (CMA) and critical process parameters (CPP) to the product CQAs; Figure 2;

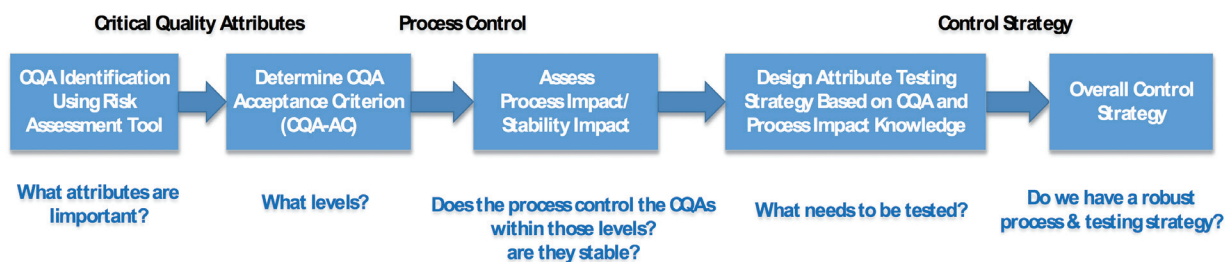


Figure 1. QbD provides a systematic approach to answer questions leading to robust process and testing strategy.

CQAs = f(CPP1, CPP2, CPP3 ...CMA1, CMA2, CMA3...)

The application of QbD to biopharmaceuticals; Quality by Design is an important element in achieving the desired state:

- Determining relationship between;
 - Quality specifications and safety or efficacy results
 - Clinical Activity and Critical Quality Attributes

- Product Attributes and Critical Process Parameters
- Process Validation and the Design Space
- Insufficient Data on “Key” versus “Critical”
- Change must be regulated

Traditional process development and validation approaches can be applied to QbD, especially in identifying CPP and defining the Design Space

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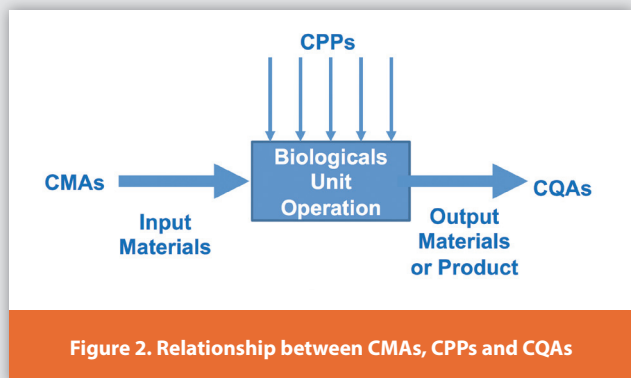
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Assessing Critical Quality Attributes

"In the case of biotechnological/biological products, most of the CQAs of the drug product are associated with the drug substance and thus are a direct result of the design of the drug substance or its manufacturing process."²

The structural and functional complexity of biotechnology proteins (e.g., Figure 3) makes identification of product CQAs challenging as a large number of attributes need to be assessed. However, at a minimum, manufacturing development should include "[Identification of] potential CQAs associated with the drug substance so that those characteristics having an impact on drug product quality can be studied and controlled"² – Expect some information at Phase I with continued refinement during development. To assess CQAs:

- Start with a list of all possible quality attributes
 - Consider mode of action and molecule type
- Risk-based approach to identify CQAs
 - Links quality attributes to safety and efficacy
 - Standardizes judgment and documents rationale
- Criticality reflects impact on safety and efficacy
- Keep process considerations separate from CQA assessment
 - CQA impact on safety and efficacy is independent of process capability, process changes shouldn't impact quality attribute criticality
 - Makes CQA assessment more modular

Approach to Identify CQAs

1. Consider all DP quality attributes; physical attributes, identification, assay, content uniformity, dissolution and drug release, degradation products, residual solvents, moisture, microbial limits, etc.
2. Identify a CQA based on the severity of harm to a patient (safety and efficacy) resulting from failure to meet that quality attribute. – Identified before taking into account risk control – Does not change as a result of risk management.

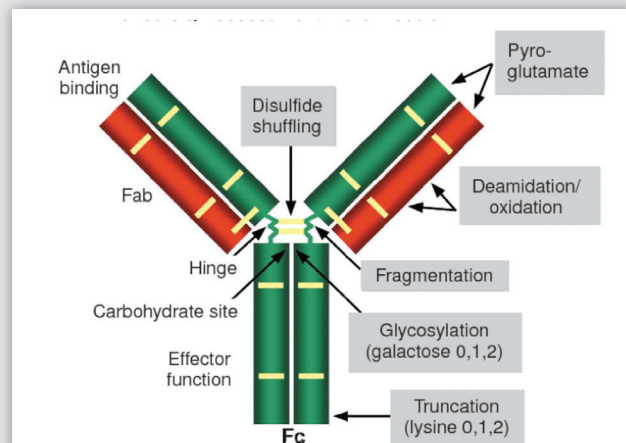
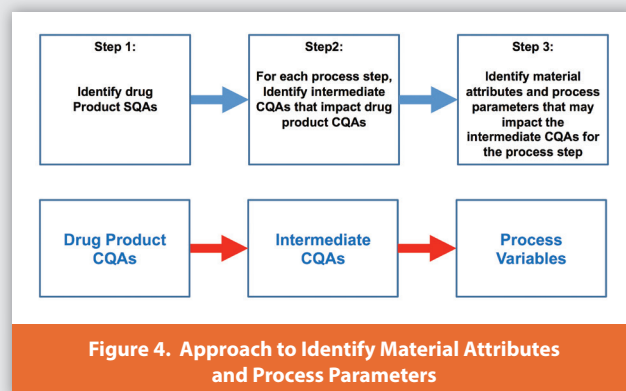


Figure 3. "Typical forms of protein heterogeneity. Protein therapeutics such as mAbs have the potential to be affected by one or more forms of heterogeneity. Both susceptibility to heterogeneity and its effect, if any, on clinical efficacy and safety is highly product-dependent. An immunoglobulin (IgG) is depicted with the heavy chain in green, the light chain in red and disulfide bridges in yellow. Antibody function potentially affected by heterogeneity, and structural features involved in giving rise to heterogeneity are listed on the left whereas different forms of heterogeneity are denoted in gray boxes. Note that although deamidation/oxidation is depicted as occurring exclusively in the Fab region, there are examples of additional susceptible residues located throughout the molecule. Glycosylation heterogeneity can lead to multiple carbohydrate species (e.g., G0, G1, G2) whereas fragmentation can be exacerbated by proteolytic susceptibility of the hinge region. Many forms of heterogeneity listed here can affect additional classes of protein products."¹³



Potential Critical Quality Attributes (pCQA's)

Critical quality attributes are linked to patient safety and efficacy. Physical, chemical, biological or microbiological properties or characteristics that should be within an appropriate limit, range or distribution to ensure the desired product quality:

- Assessed for potential impact on Safety, Immunogenicity, PK or Potency as relevant
- Criticality analysis is basis for justification to specify or not and for acceptable ranges
- CQA's cannot be deemed as less critical due to process control

As it become obvious from following QbD and CQAs definitions that CQAs is a requirement for QbD to focus on:

- "Means that product and process performance characteristics are scientifically designed to meet specific objectives, not merely empirically derived from performance of test batches."
- The product is designed to meet patient needs and performance requirements
- The process is designed to consistently meet product critical quality attributes
- The impact of starting raw materials and process parameters on product quality is well understood
- The process is continually monitored, evaluated and updated to allow for consistent quality throughout product life cycle
- Critical sources of variability are identified and controlled through appropriate control strategies

Critical Process Parameters (CPPs)

- CPPs are the independent process parameters most likely to affect the quality attributes
- Determined by sound scientific judgment and based on research, scale-up or manufacturing experience
- Controlled and monitored to confirm that the impurity profile is comparable to or better than historical data from development and manufacturing
- Quality attributes that should be considered in defining CPPs include:
 - Chemical purity
 - Qualitative and quantitative impurities
 - Physical characteristics
 - Microbial quality

The European Medicines Agency's (EMA) guideline covering "Production and Quality Control of Monoclonal Antibodies" states that "the mAb should be characterized thoroughly". "This characterization should include the determination of physicochemical and immunochemical properties, biological activity, purity, impurities, and quantity of the mAb.⁴ The EMA mAb guideline also draws attention to a number of structural features including N- and C-termini (in particular pyroglutamic acid at the N-terminus and lysine at the

C-terminus of the heavy chain), free sulfhydryl and disulphide bridge structure, glycosylation (in particular the degree of mannosylation, galactosylation, fucosylation, and sialylation), and other post-translational modifications (e.g., deamidation, oxidation, isomerisation, fragmentation, and glycation).

PAT and Biopharmaceutical Manufacturing

PAT has been defined by the United States Food and Drug Administration (FDA) as: "A mechanism to design, analyse, and control pharmaceutical manufacturing processes through the measurement of critical process parameters (CPP) which affect CQAs."⁸

Over the last decade, PAT implementation in the manufacture of small molecule pharmaceuticals is growing. This effort was driven by:⁹

- The forecasted benefits of moving from batch-based manufacturing to continuous manufacturing
- The subsequent ability to manufacture multiple active pharmaceutical ingredients (APIs) and be more responsive to changing market demand
- 'Real-time-release' (RTR), smarter manufacturing processes with feed-back and feed-forwards mechanisms/controls
- The promise of a leaner manufacturing
- Significant economic benefits

The biopharmaceutical industry has yet to realize the benefit at the production-scale. PAT and inline-online sampling/real-time analysis is still being largely research-based exercise. A transition to single-use biotechnologies, continuous bioprocessing and the implementation of PAT could afford up to 55% cost savings over a 10-year period.^{10, 11} Considering the higher costs associated with biopharmaceutical production, such savings would be significant.

Challenges of Implementing PAT

Several challenges face the manufacturing scale implementation of PAT in a bioproduction environment:

- Matrix complexity
- Analyte concentration
- Technological limitations/selectivity
- Fluidic movement
- Maintaining sterility
- Fouling of probes/sampling devices
- Obtaining cell-free samples.

Online measurements are recorded in the modern bioproduction facility, either via external monitoring (off gas/environmental), off/at-line sample analyses (extracting and taking the sample to the lab) or, more recently, via in-bioreactor mobile sensing devices.¹²

Risk Mitigation and Management

It is commonly understood that risk is defined as the combination of the probability of occurrence of harm and the severity of that harm. However, achieving a shared understanding of the application of risk management among diverse stakeholders is difficult because each stakeholder might perceive different potential of harms; place a different probability on each harm occurring and attribute different severities to each harm. In relation to biopharmaceuticals, there are a variety of stakeholders, including patients and medical practitioners as well as government and industry. The protection of the patient by managing the risk to quality should be considered of prime importance. The risk severity (consequences) and probability (likelihood it will go wrong) to the product:⁶

- What might go wrong (attribute)?
- What are the consequences (severity)?
- What is the likelihood it will go wrong (probability)?

Each quality attribute is evaluated for criticality using a risk ranking approach,⁶ which assesses the possible impact of each attribute on safety and efficacy. This ranking is determined by two factors: impact and the uncertainty (or certainty) of that impact.

The impact ranking of an attribute assesses either the known or potential consequences on safety and efficacy. The impact ranking considers the attribute's effect on:

- Efficacy, either through clinical experience or results using the most relevant potency assay(s)
- Pharmacokinetics/pharmacodynamics (PK/PD)
- Immunogenicity
- Safety

A greater understanding of the product and its manufacturing process can create a basis for more flexible regulatory approaches. The degree of regulatory flexibility is predicated on the level of relevant scientific knowledge provided in the registration application. It is the knowledge gained and submitted to the authorities, and not the volume of data collected, that forms the basis for science- and risk-based submissions and regulatory evaluations. Appropriate data demonstrating that this knowledge is based on sound scientific principles should be presented with each application. Biopharmaceutical development should include, at minimum, the following elements:

- Defining the quality target product profile (QTPP) as it relates to quality, safety and efficacy, considering e.g., the route of administration, dosage form, bioavailability, strength, and stability
- Identifying potential critical quality attributes (CQAs) of the drug product, so that those product characteristics having an impact on product quality can be studied and controlled

- Determining the critical quality attributes of the drug substance, excipients, etc., and selecting the type and amount of excipients to deliver drug product of the desired quality
- Selecting an appropriate manufacturing process
- Defining a control strategy

An enhanced, quality by design approach to product development and manufacture would additionally include the following elements:

- A systematic evaluation, understanding and refining of the formulation and manufacturing process, including
 - Identifying, through, prior knowledge, experimentation, and risk assessment, the material attributes and process parameters that can have an effect on product CQAs
 - Determining the functional relationships that link material attributes and process parameters to product CQAs
- Using the enhanced product and process understanding in combination with quality risk management to establish an appropriate control strategy, which can include a proposal for a design space(s) and/or real-time release testing

As a result, this systematic approach could facilitate continual improvement and innovation throughout the product lifecycle.⁷

Conclusion

Continuous manufacturing clearly presents significant opportunities, not only from a business perspective but from a regulatory compliance perspective. Product, process understanding and a robust control strategy will be key to successful implementations. Regulators are open to new approaches in manufacturing and control strategies; however, there is a lack of experience with continuous manufacturing models in biotech. It will be critical to engage regulators early and keep an open dialogue during the development phase.

Innovation in real-time measurement and control technologies is needed. Deeper process knowledge is required; just-in-time product disposition, less equipment, smaller portable equipment, fewer steps, faster cycle times; all are Enablers of Continuous Manufacturing to make it successfully sought and implemented.

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Author Biography



Robert Dream is an accomplished industry leader with broad experience in the Biopharmaceutical industry from early process development to commercial manufacturing. Product experience includes antibodies, vaccines, recombinant proteins, mAb's, and peptides.

Dream has been working in depth in the biopharmaceutical industry for 30 years and is involved in all aspects of manufacturing, regulatory, process validation, financial planning, development of drug substance and drug products, and technology transfer (internal and external). Dream has designed and qualified projects on all scales around the globe.

He has assisted companies to expand their operations and businesses worldwide including audit and gap analysis and assisted clients on their BLAs/NDAs preparations and submittals. He has managed contracts and improved CAPEX and OPEX on many projects, and further provided oversight, procurement and program management for clients internationally.